

Supporting Information

Cancer-ACPNet: A Two-Stage ESM-2 and Capsule Network Framework for Anticancer Peptide Screening and Cancer-Type Activity Prediction

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Supplementary TEXT S1

TEXT S1

All Stage-1 experiments were performed on a single NVIDIA GeForce RTX 5060 GPU with 24 GB GPU memory. The esm2_t30_150M_UR50D backbone was fully fine-tuned, and the representative per-fold wall-clock training times were 163.59 s for Set 1, 155.21 s for Set 2, and 85.26 s for Set 3.

Supplementary TABLE S1

TABLE S1
TABULAR REPRESENTATION OF EXISTING STUDIES ON ACP AND ACP ACTIVITY TYPE CLASSIFICATION.

Classifier Name	Operating Algorithm	Publication Year
Multi-ACPNet [15]	BiLSTM, CNN & GCN	2026
ACP-CLB [6]	CNN, BiLSTM, ProtBERT + Conv1D	2025
DUO-ACP [8]	Pre-training-based model	2025
ACP-CapsPred [7]	Capsule Network	2024
ACP-LSTM-NFR [10]	LSTM	2024
ACP-PDAFF [13]	Pre-training-based model (ProtBert)	2024
MA-PEP [14]	Attention-based multimodal	2024
mACPpred2.0 [16]	Stacked Deep Learning (SDL)	2024
ACP-LSE [17]	Deep neural network	2024
ACP-MLC [9]	Random Forest	2023
GRDF [11]	Deep Forest	2023
TriNet [12]	CNN, Bi-LSTM & Transformer encoder	2023
ACP-OPE [20]	BiLSTM, CNN & LightGBM	2023
Me-ACP [19]	Hybrid CNN+LSTM deep neural network	2022
iACP-DRLF [18]	LSTM + LightGBM	2021

Supplementary TABLE S2

TABLE S2
DISTRIBUTION OF TRAINING AND TESTING ON SET 1, SET 2, AND SET 3 DATASETS

Dataset	Train/Test	Positive	Negative	Total
Set 1	Train dataset	632	641	1273
	Test dataset	161	158	319
Set 2	Train dataset	716	677	1393
	Test dataset	186	170	356
Set 3	Train dataset	242	242	484
	Test dataset	61	61	122

Supplementary TABLE S3

TABLE S3
DISTRIBUTION OF THE STAGE-2 DATASET USED FOR ONE-VS-ALL CANCER-TYPE PREDICTION

Cancer Types	No. of ACPs	Training Positive/Negative	Testing Positive/Negative
Colon	137	109/59	28/15
Breast	124	99/69	25/18
Cervix	105	84/84	21/22
Skin	106	84/84	22/21
Lung	94	75/93	19/24
Prostate	89	71/97	18/25
Blood	63	50/118	13/30

Supplementary TABLE S4

TABLE S4

PERFORMANCE COMPARISON OF THE PROPOSED MODEL AND EXISTING METHODS FOR ACP FUNCTIONAL ACTIVITY PREDICTION ACROSS SEVEN CANCER TYPES UNDER THE REDUCED 5-CLASS BENCHMARK SETTING.

Cancer type	Method	ACC	Precision	Recall	F1-score	MCC
Colon	ACP-CapsPred	91.60%	90.83%	92.16%	91.49%	83.17%
	ACP-MLC	91.89%	96.07%	92.45%	94.23%	87.44%
	Proposed Model	87.50%	92.59%	89.29%	90.91%	71.05%
Breast	ACP-CapsPred	94.59%	96.08%	96.08%	96.08%	84.55%
	ACP-MLC	93.33%	92.12%	92.72%	92.42%	78.22%
	Proposed Model	95.00%	100.00%	92.00%	95.83%	90.10%
Cervix	ACP-CapsPred	94.59%	92.11%	97.22%	94.60%	91.93%
	ACP-MLC	49.31%	51.35%	50.00%	50.66%	83.77%
	Proposed Model	92.50%	90.91%	95.24%	93.02%	85.03%
Skin	ACP-CapsPred	90.54%	90.00%	92.31%	91.14%	81.03%
	ACP-MLC	71.16%	67.00%	84.47%	74.73%	45.75%
	Proposed Model	95.00%	90.48%	100.00%	95.00%	90.48%
Lung	ACP-CapsPred	82.43%	85.14%	82.05%	83.57%	70.55%
	ACP-MLC	55.55%	55.55%	100.00%	71.42%	28.25%
	Proposed Model	92.50%	90.91%	95.24%	93.02%	85.84%
Average	ACP-CapsPred	90.75%	90.83%	91.96%	91.38%	82.24%
	ACP-MLC	72.25%	72.42%	83.93%	76.69%	64.69%
	Proposed Model	92.50%	92.98%	94.35%	93.56%	84.50%

Supplementary TABLE S5

TABLE S5

COMPARISON OF FIRST-TOKEN, MEAN, MAX, AND CONCATENATED FIRST-TOKEN AND MEAN POOLING FOR CANCER-ACPNET ON SET 1, SET 2, AND SET 3.

Dataset	Pooling Strategy	Accuracy	Precision	F1-Score	MCC	AUC
Set 1	Mean pooling	82.07%	88.69%	81.48%	64.45%	87.12%
	Max pooling	82.95%	87.61%	82.09%	66.27%	87.13%
	First & Mean pooling	83.57%	89.90%	83.08%	67.34%	88.42%
	Cancer-ACPNet(First)	84.08%	84.62%	83.95%	68.22%	89.51%
Set 2	Mean pooling	95.96%	98.33%	96.02%	92.06%	97.22%
	Max pooling	95.39%	98.76%	95.47%	90.92%	98.31%
	First & Mean pooling	96.74%	98.81%	96.82%	93.57%	98.07%
	Cancer-ACPNet(First)	96.74%	98.99%	96.81%	93.59%	97.27%
Set 3	Mean pooling	82.46%	86.14%	82.73%	65.18%	87.33%
	Max pooling	84.75%	85.59%	85.53%	70.40%	88.33%
	First & Mean pooling	83.93%	86.72%	84.81%	68.54%	88.18%
	Cancer-ACPNet(First)	84.75%	81.82%	85.39%	70.04%	88.36%

Supplementary TABLE S6

TABLE S6
STAGE-1 CALIBRATION RESULTS BEFORE AND AFTER TEMPERATURE SCALING

Dataset	Temperature	ECE_before	ECE_after	Brier_before	Brier_after	NLL_before	NLL_after
SET 1	0.5189	0.1591	0.0399	0.1451	0.1206	0.4704	0.4002
SET 2	0.2484	0.2304	0.0221	0.0892	0.0333	0.3429	0.1620
SET 3	0.1406	0.2610	0.0489	0.2006	0.1338	0.5931	0.4365

Supplementary TABLE S7

TABLE S7
QUANTITATIVE SALIENCY STABILITY AND FDR-CORRECTED RESIDUE ENRICHMENT RESULTS

ANALYSIS SUBSET	N	BOOTSTRAP REPLICATES	MEAN SPEARMAN	MEDIAN SPEARMAN	SPEARMAN 5TH-95TH PERCENTILE	MEAN TOP-BIN JACCARD	MEDIAN TOP-BIN JACCARD	JACCARD 5TH-95TH PERCENTILE
ACP- POSITIVE VALIDATION PEPTIDES	200	200	0.652	0.740	0.181-0.984	0.361	0.250	0.111-0.667
NON-ACP VALIDATION PEPTIDES	200	200	0.658	0.764	0.173-0.975	0.509	0.429	0.250-1.000

Supplementary TABLE S8

TABLE S8

PERFORMANCE COMPARISON OF ALTERNATIVE PCA AND MLP DIMENSIONALITY SETTINGS FOR STAGE-1 CLASSIFICATION ON SET 1, SET 2, AND SET 3.

Dataset	Methodology	ACC	Precision	Recall	F1-Score	MCC	AUC
Set 1	pca128_mlp128	0.8332	0.8454	0.8186	0.8313	0.6675	0.8822
	pca256_mlp64	0.8401	0.8547	0.8199	0.8362	0.6817	0.8899
	pca256_mlp256	0.8188	0.8446	0.7826	0.8111	0.6411	0.8610
	pca384_mlp128	0.8351	0.8437	0.8224	0.8322	0.6713	0.8808
	pca256_mlp128 (Proposed)	0.8408	0.8462	0.8335	0.8395	0.6822	0.8951
Set 2	pca128_mlp128	0.9601	0.9798	0.9430	0.9610	0.9210	0.9701
	pca256_mlp64	0.9691	0.9856	0.9548	0.9699	0.9387	0.9787
	pca256_mlp256	0.9663	0.9844	0.9505	0.9671	0.9332	0.9793
	pca384_mlp128	0.9652	0.9876	0.9452	0.9659	0.9313	0.9807
	pca256_mlp128 (Proposed)	0.9674	0.9899	0.9473	0.9681	0.9359	0.9727
Set 3	pca128_mlp128	0.8361	0.8296	0.8459	0.8369	0.6739	0.8947
	pca256_mlp64	0.8246	0.7950	0.8885	0.8369	0.6582	0.8787
	pca256_mlp256	0.8301	0.8016	0.8807	0.8392	0.6636	0.8741
	pca384_mlp128	0.8393	0.8124	0.8820	0.8456	0.6816	0.8799
	pca256_mlp128 (Proposed)	0.8475	0.8182	0.8951	0.8539	0.7004	0.8836

Supplementary TABLE S9

TABLE S9
PERFORMANCE COMPARISON OF ALTERNATIVE PCA AND MLP DIMENSIONALITY SETTINGS FOR STAGE-1
CLASSIFICATION ON SET 1, SET 2, AND SET 3.

Class	Configuration	Accuracy	Sensitivity	Specificity	F1-score	AUC	MCC	Δ Accuracy
Breast	Complete model	0.9302	0.9600	0.8889	0.9412	0.9378	0.8565	Reference
	Without balancing	0.9070	0.9200	0.8889	0.9200	0.9556	0.8089	−0.0232
	Without MixUp	0.8140	0.9200	0.6667	0.8519	0.9356	0.6176	−0.1162
	Without focal loss (BCE)	0.8372	0.7600	0.9444	0.8444	0.9467	0.6967	−0.0930
Blood	Complete model	0.9070	0.7692	0.9667	0.8333	0.9308	0.7746	Reference
	Without balancing	0.9070	0.8462	0.9333	0.8462	0.9538	0.7795	0.0000
	Without MixUp	0.8372	0.8462	0.8333	0.7586	0.8974	0.6456	−0.0698
	Without focal loss (BCE)	0.7442	0.7692	0.7333	0.6452	0.9077	0.4679	−0.1628

Supplementary TABLE S10

TABLE S10
PERFORMANCE OF CANCER-ACPNET ON SET 1, SET 2, AND SET 3 WITH 95% STRATIFIED BOOTSTRAP
CONFIDENCE INTERVALS.

Dataset	ACC, (95% CI)	Precision, (95% CI)	Recall, (95% CI)	F1-score, (95% CI)	MCC, (95% CI)	FPR, (95% CI)	AUC, (95% CI)
Set 1	84.08% (82.26– 85.83)	84.62% (82.55– 86.82)	83.35% (80.87– 85.71)	83.95% (82.00– 85.75)	68.22% (64.65– 71.79)	15.19% (12.66– 17.59)	89.51% (87.84– 91.02)
Set 2	96.74% (95.84– 97.58)	98.99% (98.32– 99.56)	94.73% (93.23– 96.13)	96.81% (95.91– 97.63)	93.59% (91.90– 95.20)	1.06% (0.47–1.76)	97.27% (96.36– 98.10)
Set 3	84.75% (81.96– 87.54)	81.82% (78.60– 85.48)	89.51% (86.23– 92.79)	85.39% (82.72– 88.05)	70.04% (64.59– 75.80)	20.00% (15.74– 24.26)	88.36% (85.68– 91.14)

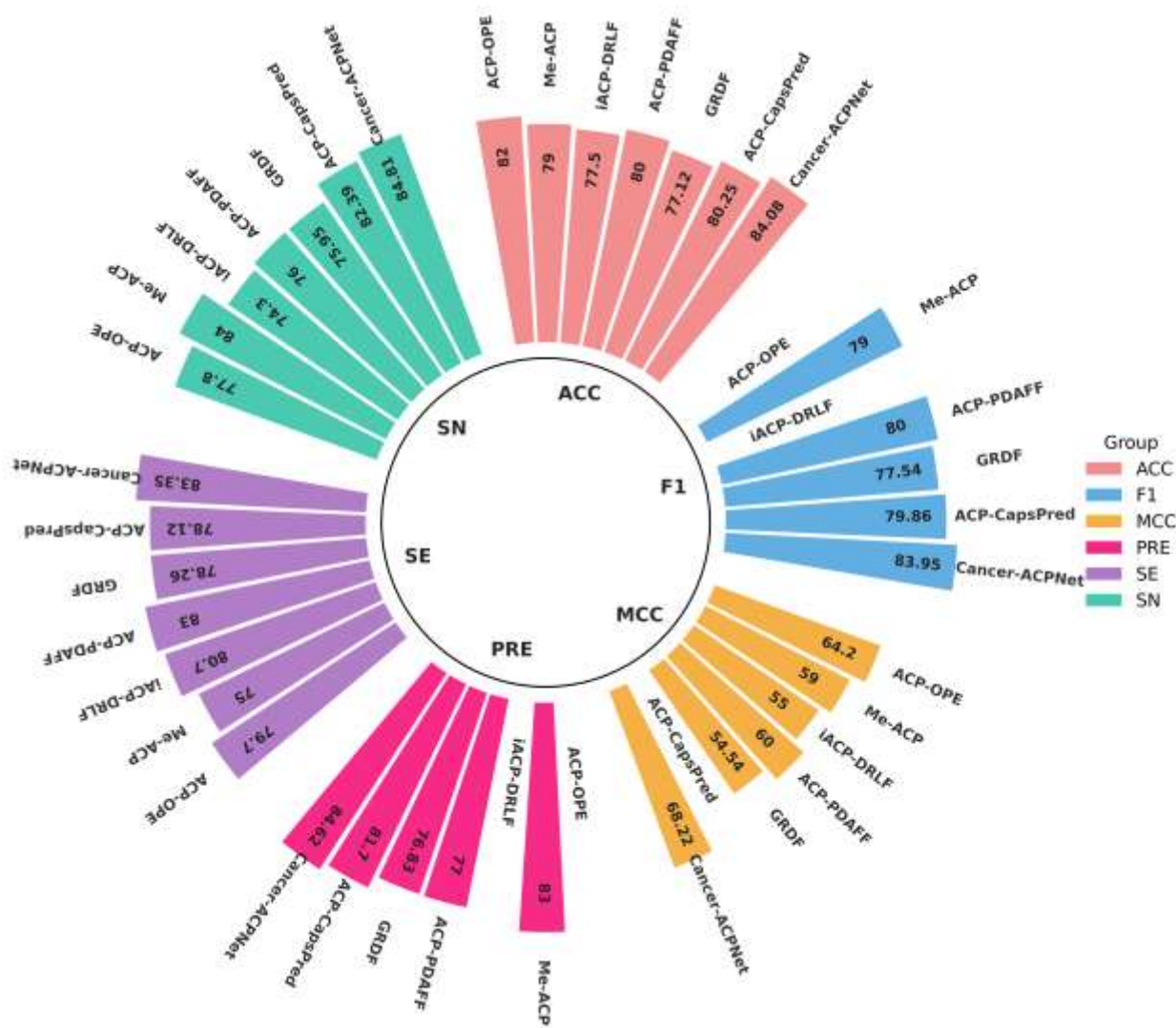
Supplementary TABLE S11

TABLE S11
**CANCER-TYPE ACTIVITY PREDICTION PERFORMANCE OF CANCER-ACPNET ACROSS SEVEN ONE-
 VS-ALL TASKS WITH 95% STRATIFIED BOOTSTRAP CONFIDENCE INTERVALS.**

Cancer type	ACC, (95% CI)	SE, (95% CI)	SP, (95% CI)	F1-score, (95% CI)	AUC, (95% CI)	MCC, (95% CI)
Colon	93.02% (83.72–100.00)	96.43% (89.29–100.00)	86.67% (66.67–100.00)	94.74% (88.52–100.00)	96.67% (87.14–100.00)	84.52% (65.31–100.00)
Breast	93.02% (83.72–100.00)	96.00% (88.00–100.00)	88.89% (72.22–100.00)	94.12% (87.27–100.00)	93.78% (83.11–100.00)	85.65% (67.97–100.00)
Cervix	83.72% (72.09–93.02)	86.36% (72.73–100.00)	80.95% (61.90–95.24)	84.44% (73.17–93.62)	89.39% (76.62–96.75)	67.46% (44.23–86.85)
Skin	86.05% (74.42–95.35)	78.95% (57.89–94.74)	91.67% (79.17–100.00)	83.33% (68.75–94.44)	89.91% (76.10–98.25)	71.72% (49.03–90.57)
Lung	88.37% (79.07–97.67)	86.36% (72.73–100.00)	90.48% (76.19–100.00)	88.37% (77.27–97.67)	94.81% (84.20–100.00)	76.84% (58.13–95.44)
Prostate	90.70% (81.40–97.67)	88.89% (72.22–100.00)	92.00% (80.00–100.00)	88.89% (77.42–97.30)	91.78% (86.89–99.56)	80.89% (61.78–95.37)
Blood	90.70 (81.40–97.67)	76.92 (53.85–100.00)	96.67 (90.00–100.00)	83.33 (63.64–96.00)	93.08 (87.44–98.97)	77.46 (54.25–94.51)
Macro-average	89.37% (85.71–92.69)	87.13% (81.19–92.56)	89.62% (84.38–94.35)	88.18% (83.76–91.98)	92.77% (89.15–95.21)	77.79% (70.68–84.97)

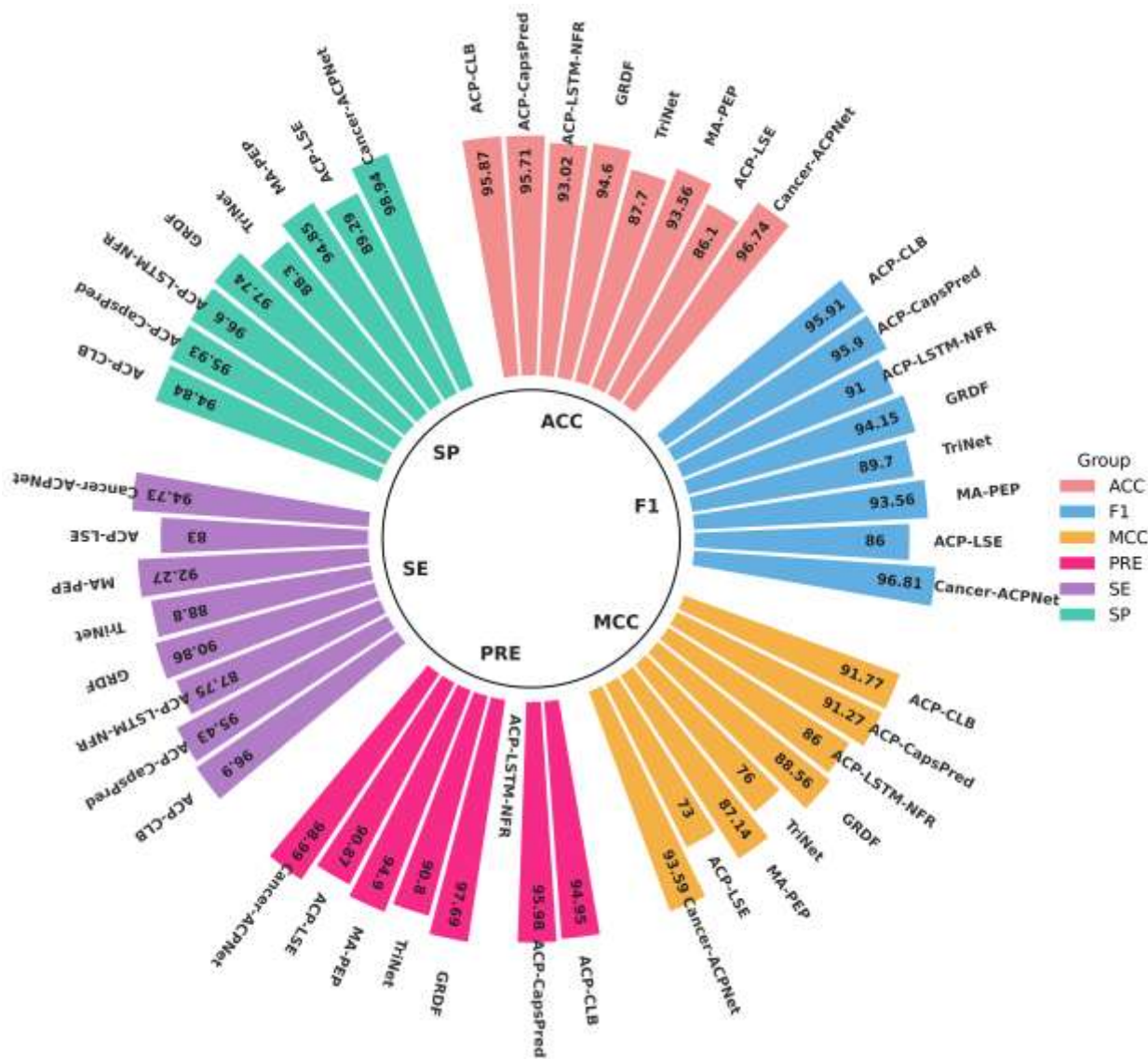
Supplementary Fig. S1

(A)



Supplementary Fig. S1

(B)



Supplementary Fig. S1

(C)

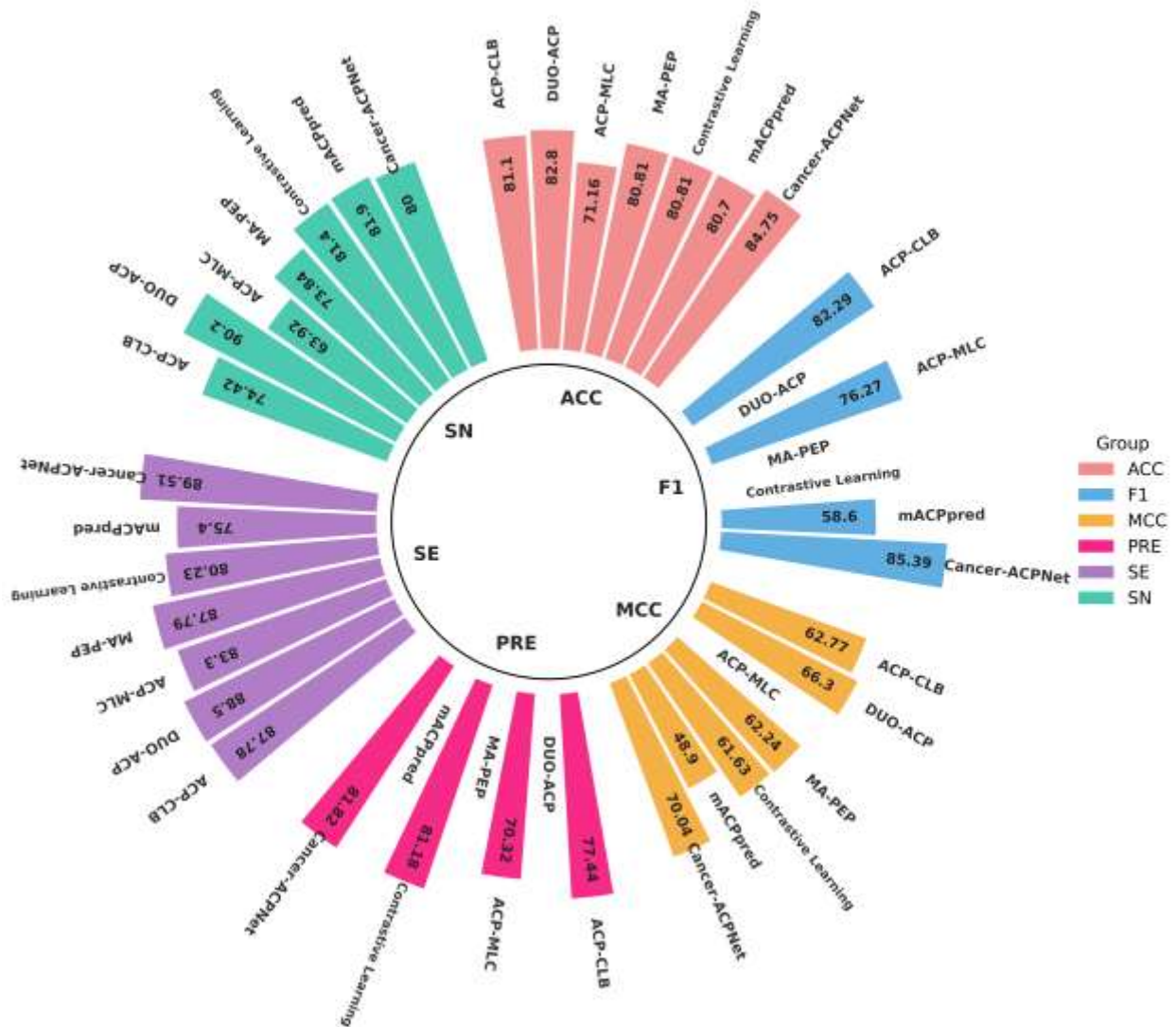
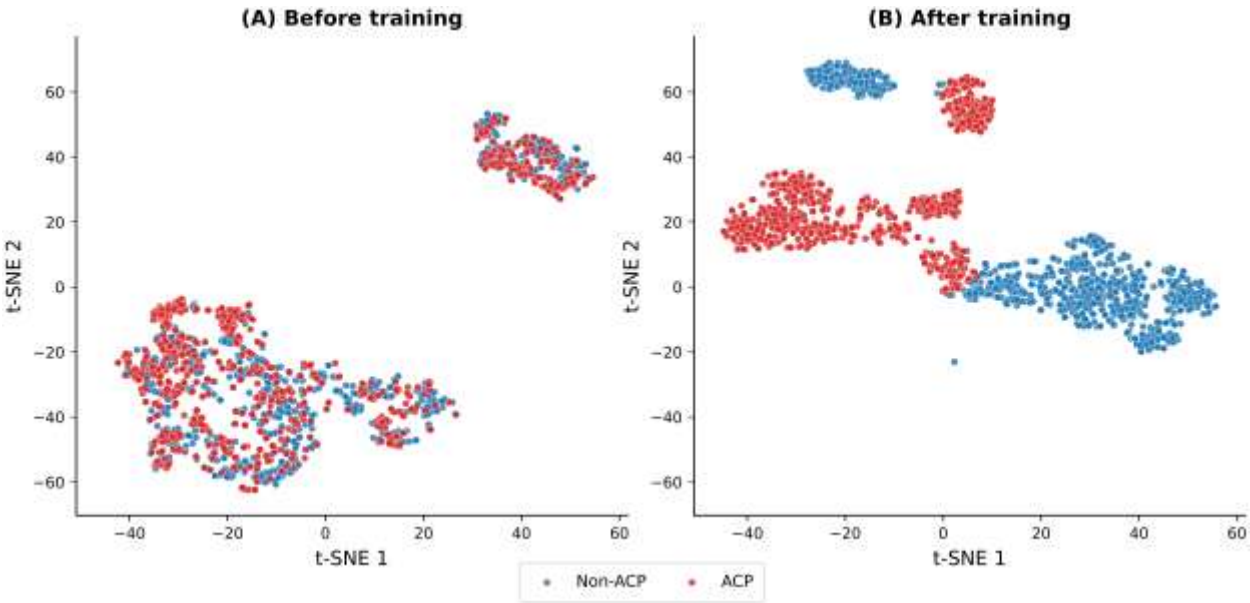


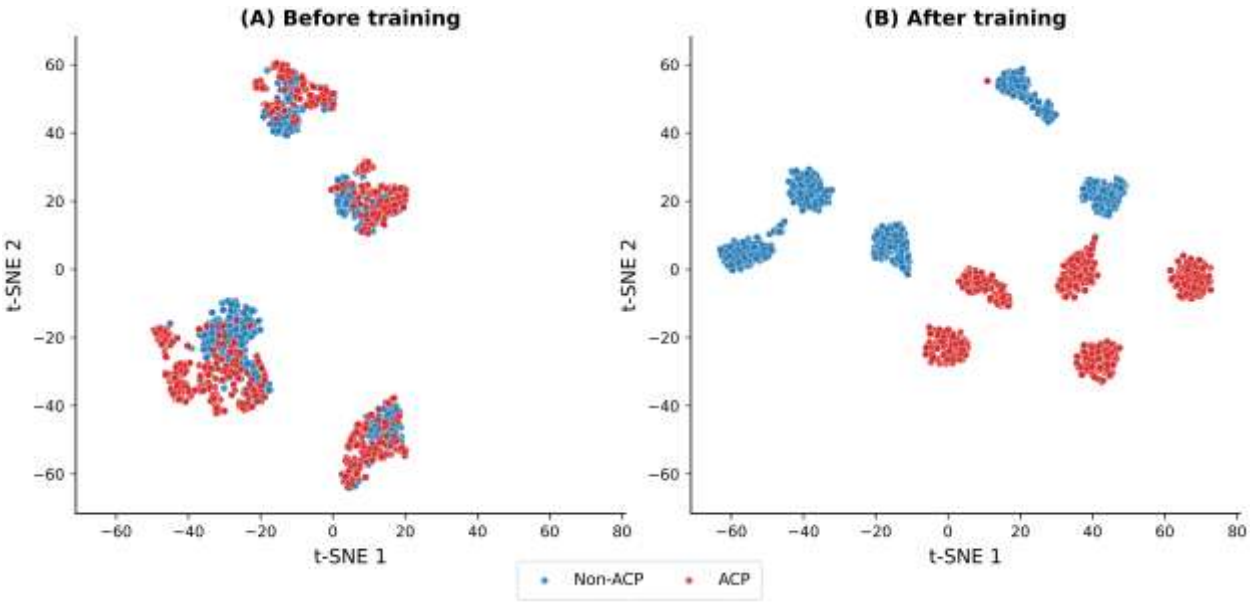
Fig. S1. Comparison of the proposed model with previously reported ACP prediction methods on (A) Set 1, (B) Set 3, across six evaluation metrics: ACC, PRE, SE, SP, F1-score, and MCC.

Supplementary Fig. S2

(A)



(B)



Supplementary Fig. S2

(C)

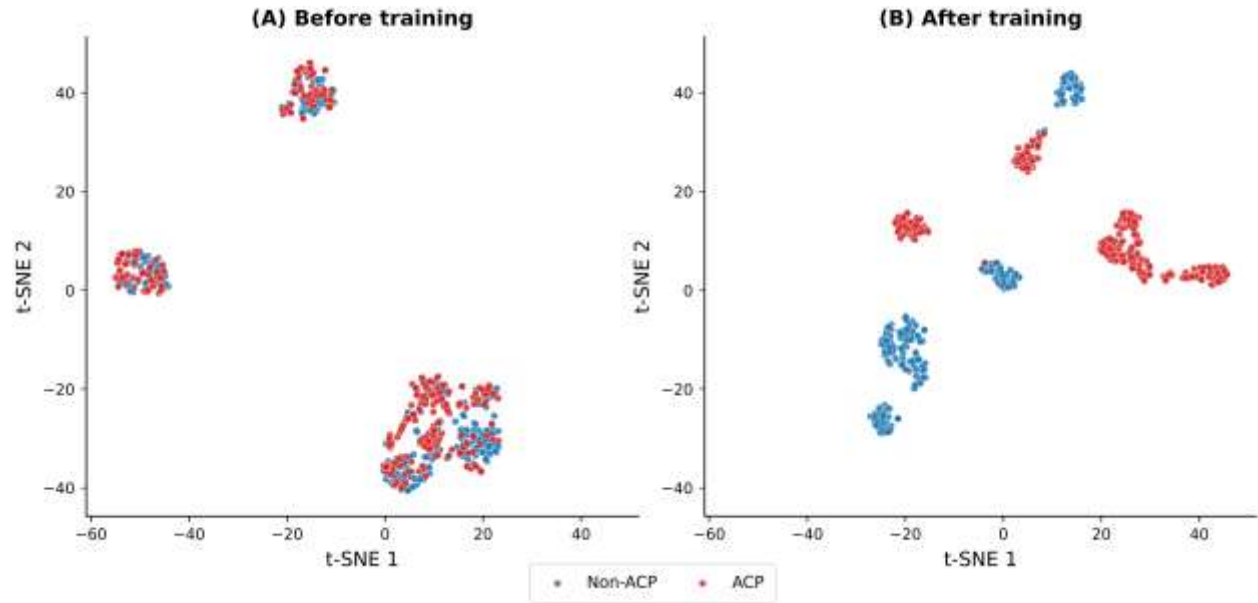


Fig. S2. t-SNE visualization of the Stage-1 capsule representations. Panels (A), (B), and (C) show Set 1, Set 2, and Set 3, respectively. Each panel compares the 32-dimensional digit-capsule features before training, colored by true labels, with those after training, colored by predicted labels. The plots were generated using a perplexity of 30, 1,000 iterations, and a random seed of 42 and are provided only as qualitative visualizations of the feature space.

Supplementary Fig. S3

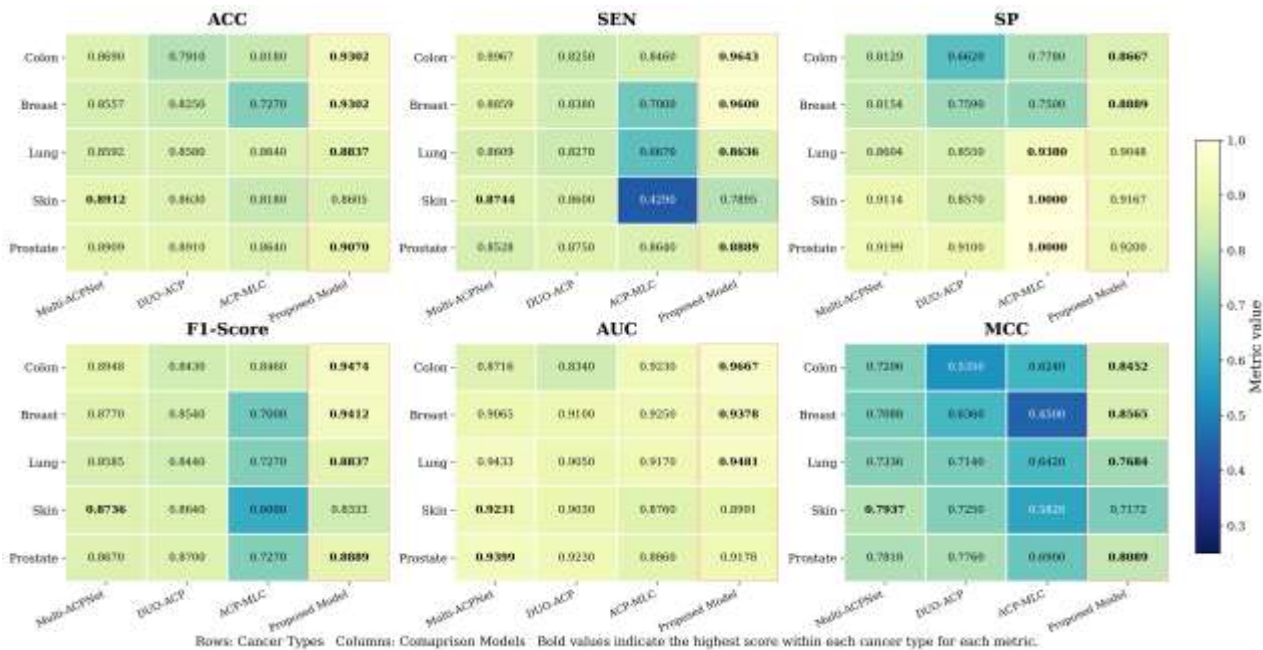
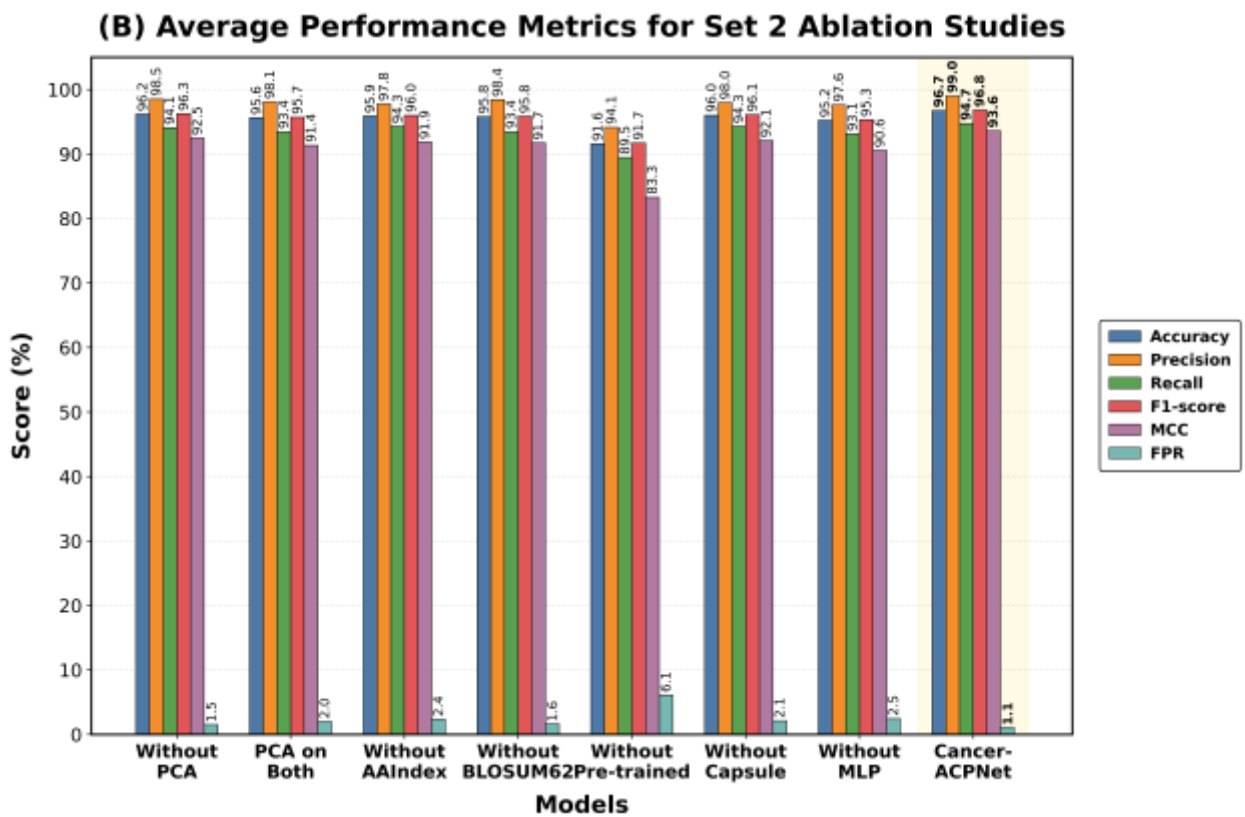
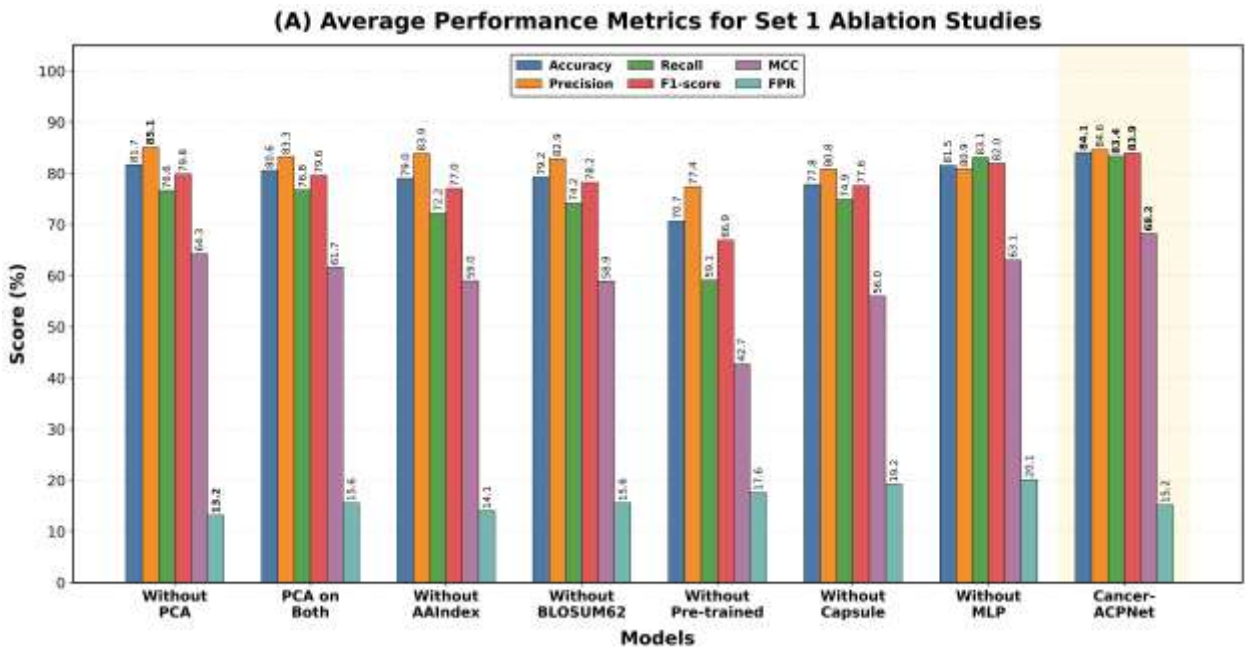


Fig. S3. Heatmap visualization of class-wise ACP functional activity prediction results under the reduced 5-class benchmark. The figure compares Multi-ACPNNet, DUO-ACP, ACP-MLC, and the proposed model across five cancer types using six evaluation metrics: ACC, SEN, SP, F1-score, AUC, and MCC. Bold entries denote the best-performing model for each cancer type and metric.

Supplementary Fig. S4



Supplementary Fig. S4

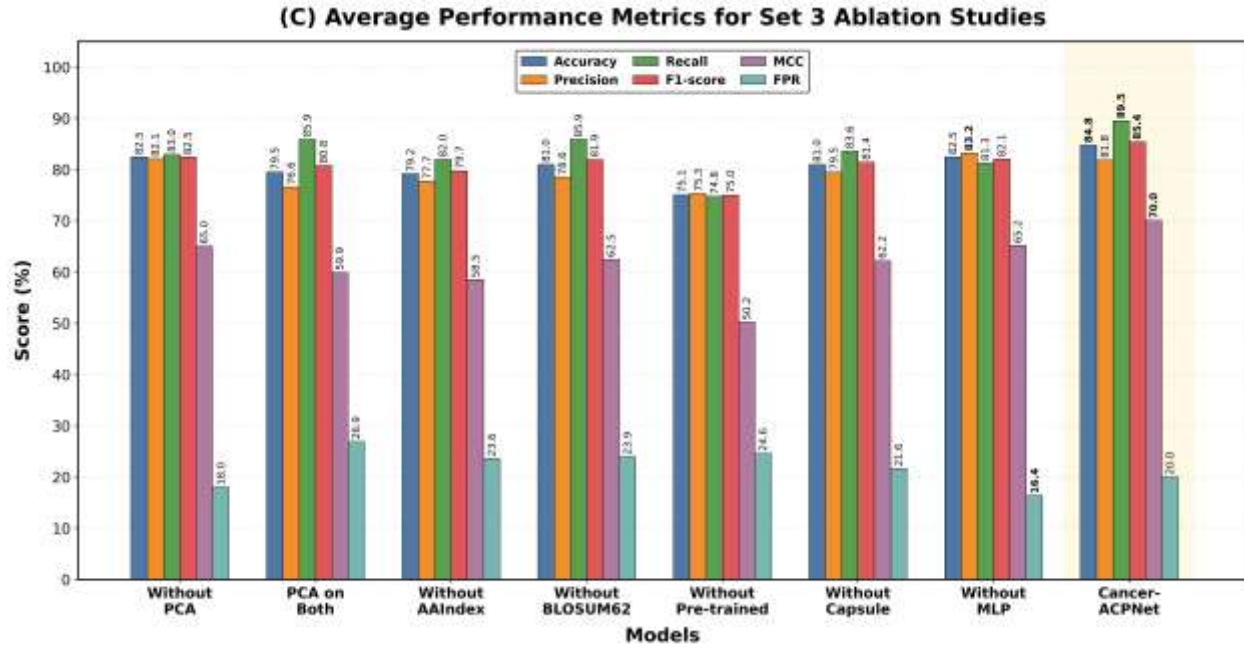


Fig. S4. Visual comparison of the (A) ablation study results on Set 1, (B) ablation study results on Set 2, (C) ablation study results on Set 3. The bar plot summarizes the average performance of the proposed model and its ablated variants in terms of ACC, Precision, Recall, F1-score, MCC, and FPR. The “Without Pre-trained Model” variant removes the ESM-2 branch completely and retains only the PCA-compressed BLOSUM62/AAIndex feature branch, capsule layers, and MLP classifier.

Supplementary Fig. S5

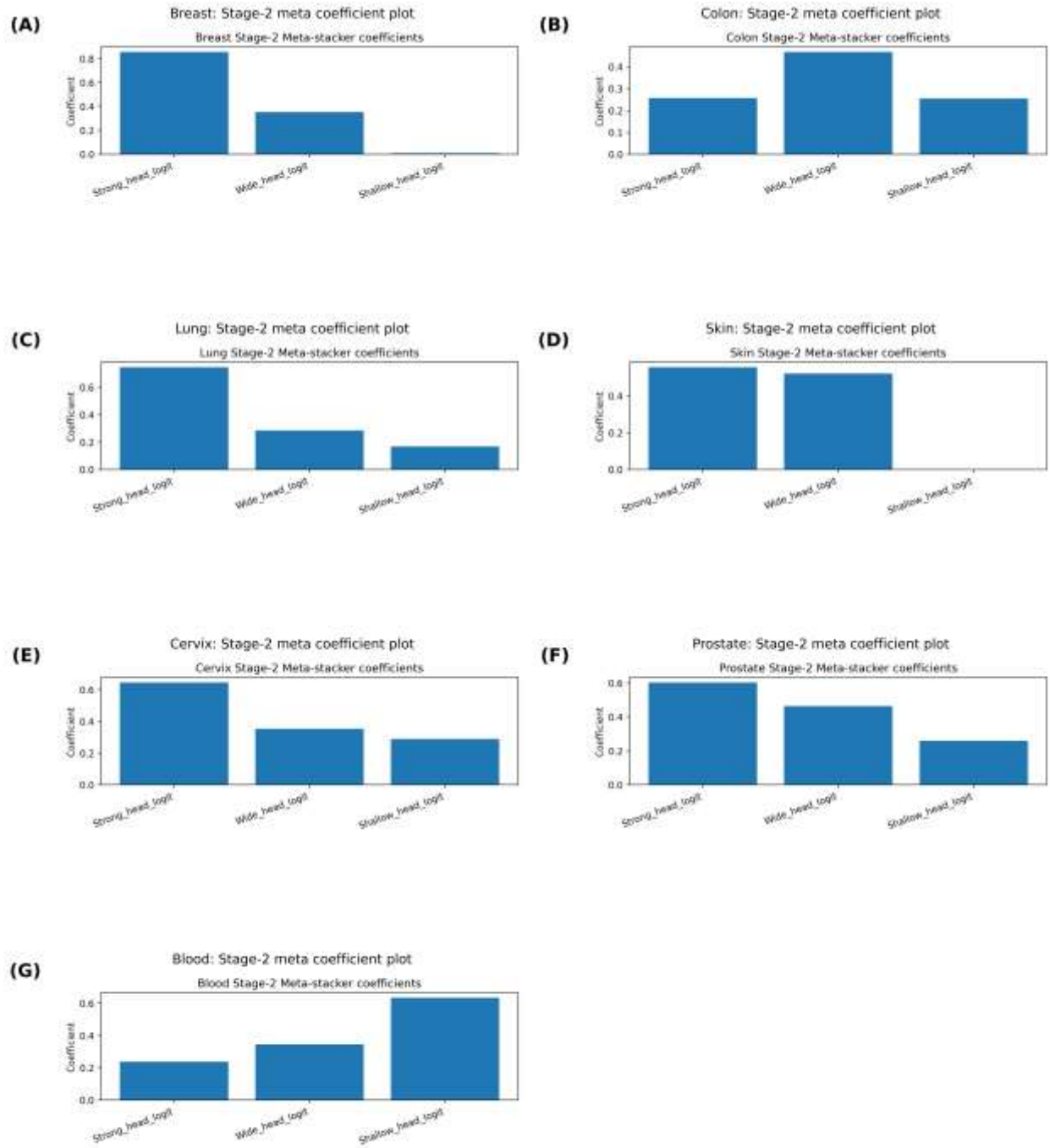


Fig. S5. Meta-stacker coefficients for the Strong, Wide, and Shallow Stage-2 heads across the seven one-vs-all cancer-type models. These coefficients indicate how much each prediction head contributed to the final meta-stacked decision for each cancer type (A-G); they are model-combination weights, not evaluation metrics such as accuracy, AUC, or MCC.

Supplementary Fig. S6

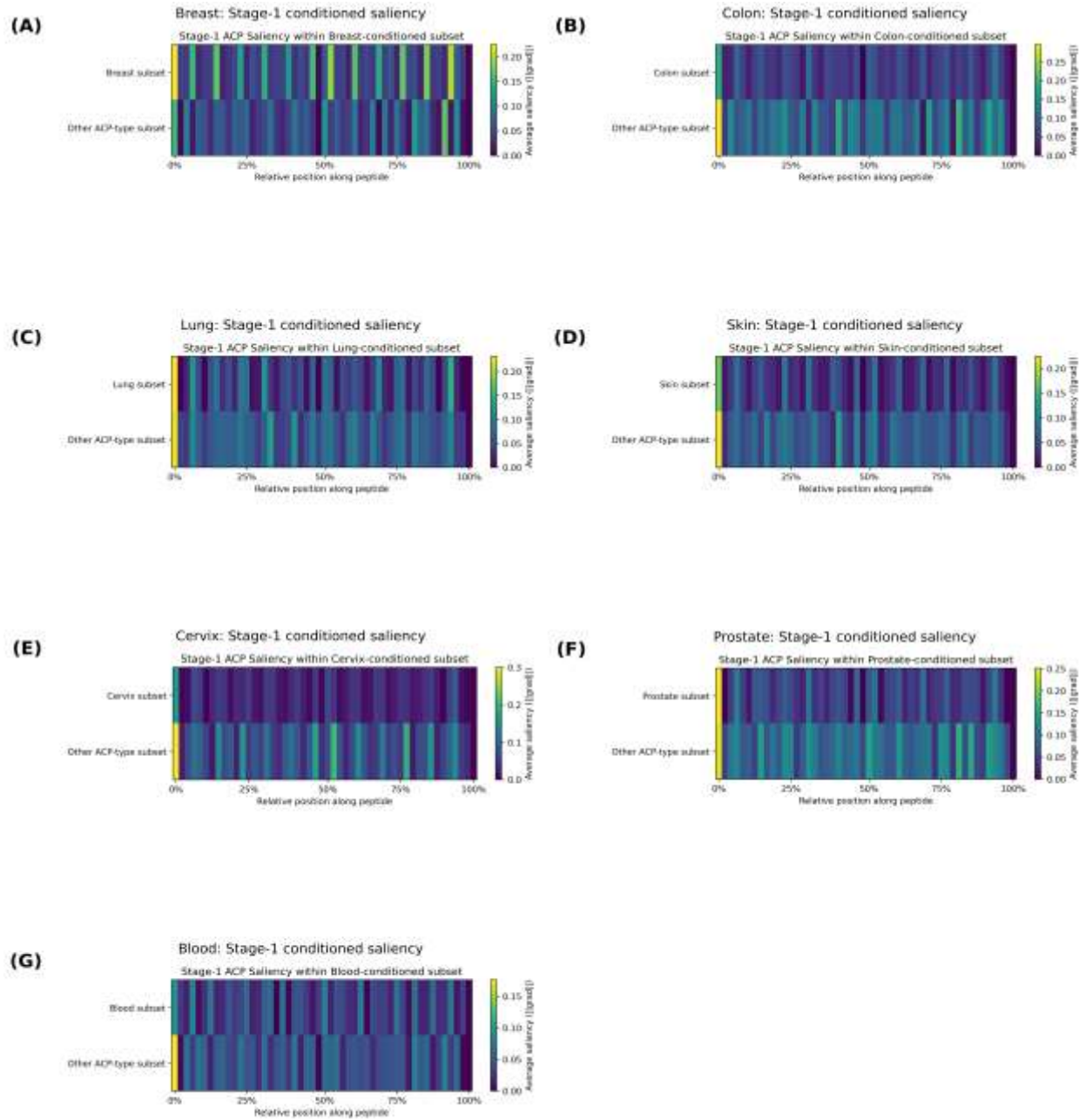


Fig. S6. Average Stage-1 gradient saliency profiles for cancer-type-conditioned peptide subsets from A-G. Each panel compares peptides belonging to the indicated cancer type with other ACP-type peptides across normalized peptide positions from 0% to 100%. Brighter regions indicate positions with higher average saliency for the Stage-1 ACP screening model.

Supplementary Fig. S7

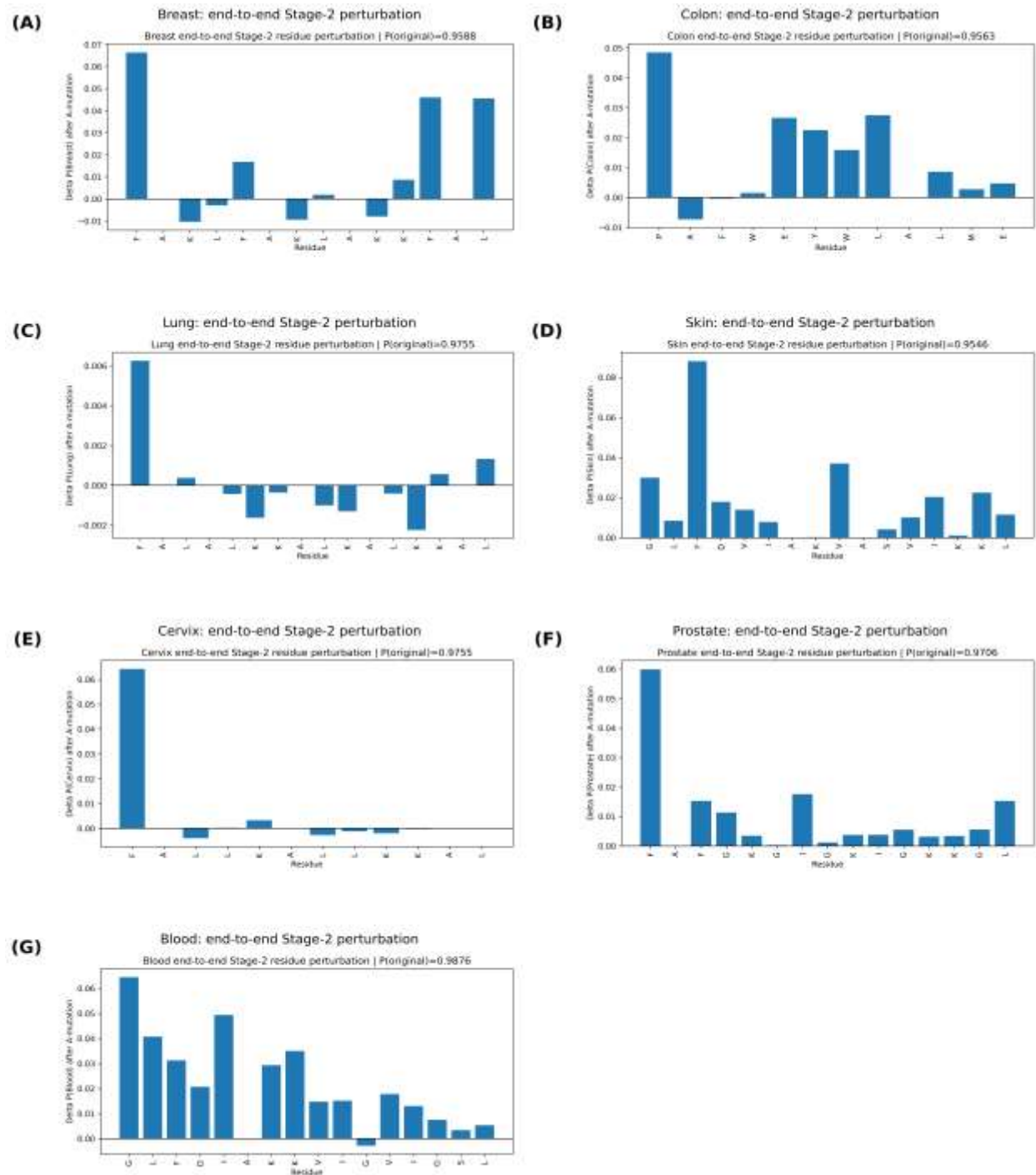


Fig. S7. Representative end-to-end perturbation examples for the seven cancer-type activity models from A-G. Each panel shows the change in the predicted probability for the corresponding cancer type after mutating individual peptide residues. Larger positive values indicate residues whose mutation produced a stronger decrease in the original cancer-type prediction.

Supplementary Fig. S8

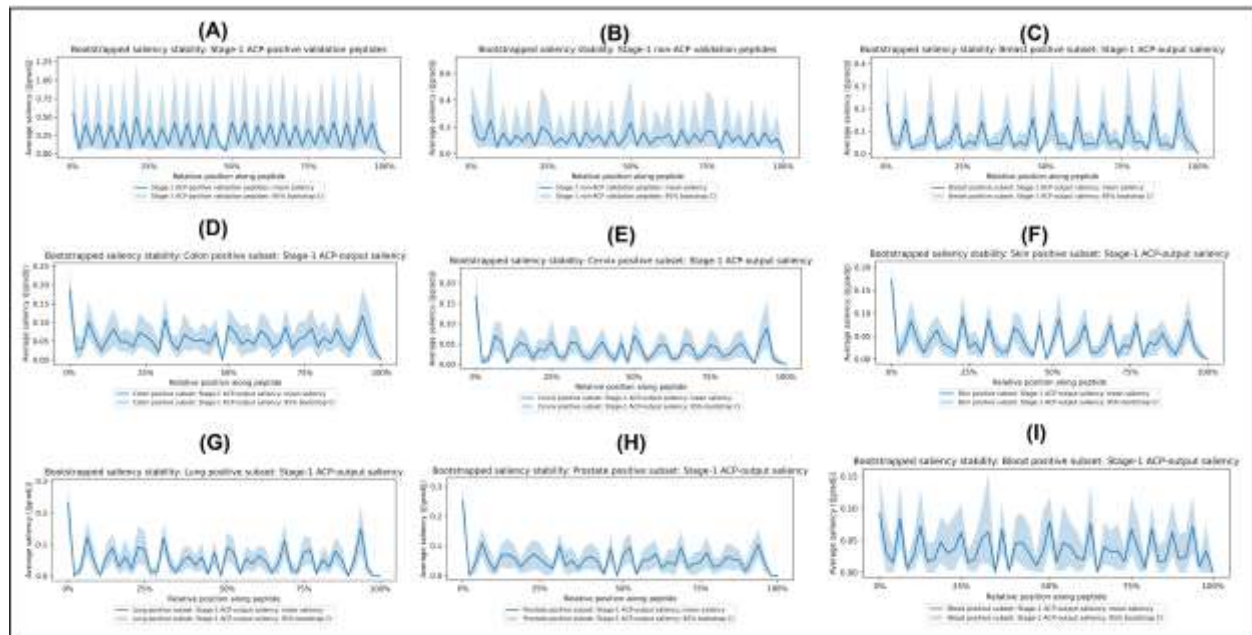
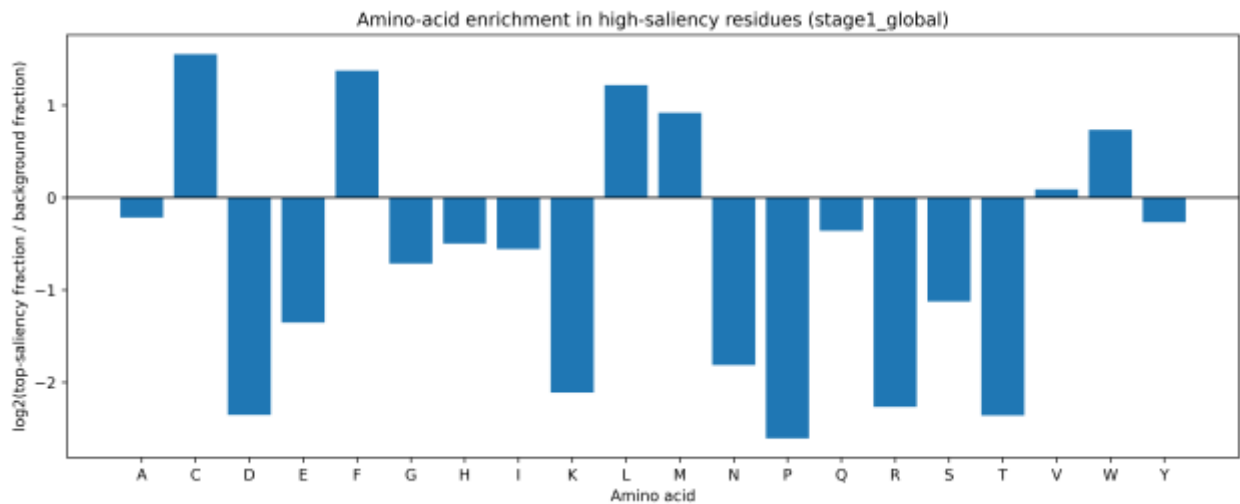


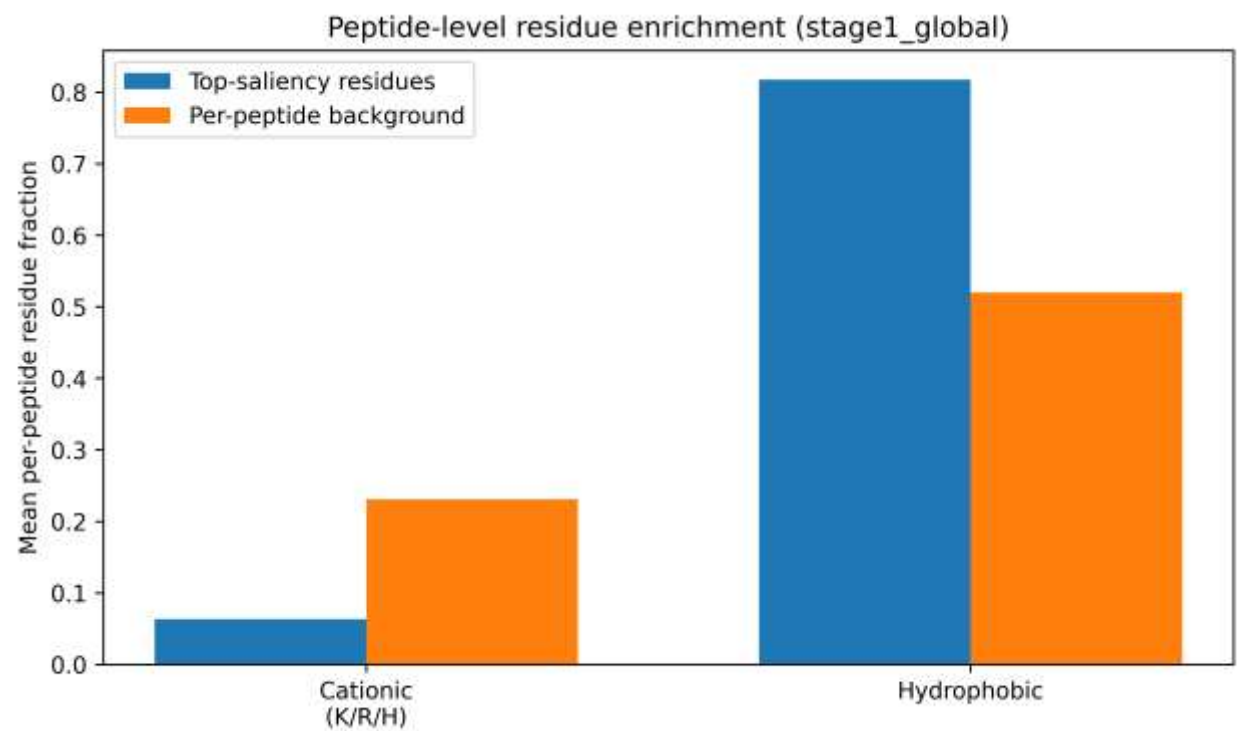
Fig. S8. Bootstrap stability of the Stage-1 saliency profiles. Mean residue-position saliency and 95% percentile bootstrap intervals were obtained from 200 peptide-level bootstrap replicates. Panel (A-B) compares the global ACP and non-ACP profiles separately, and the remaining panels (C-I) show Stage-1 ACP-output saliency within the seven cancer-type subsets. These results assess the resampling stability of the model explanations and do not establish biological mechanisms.

Supplementary Fig. S9

(A)



(B)



Supplementary Fig. S9

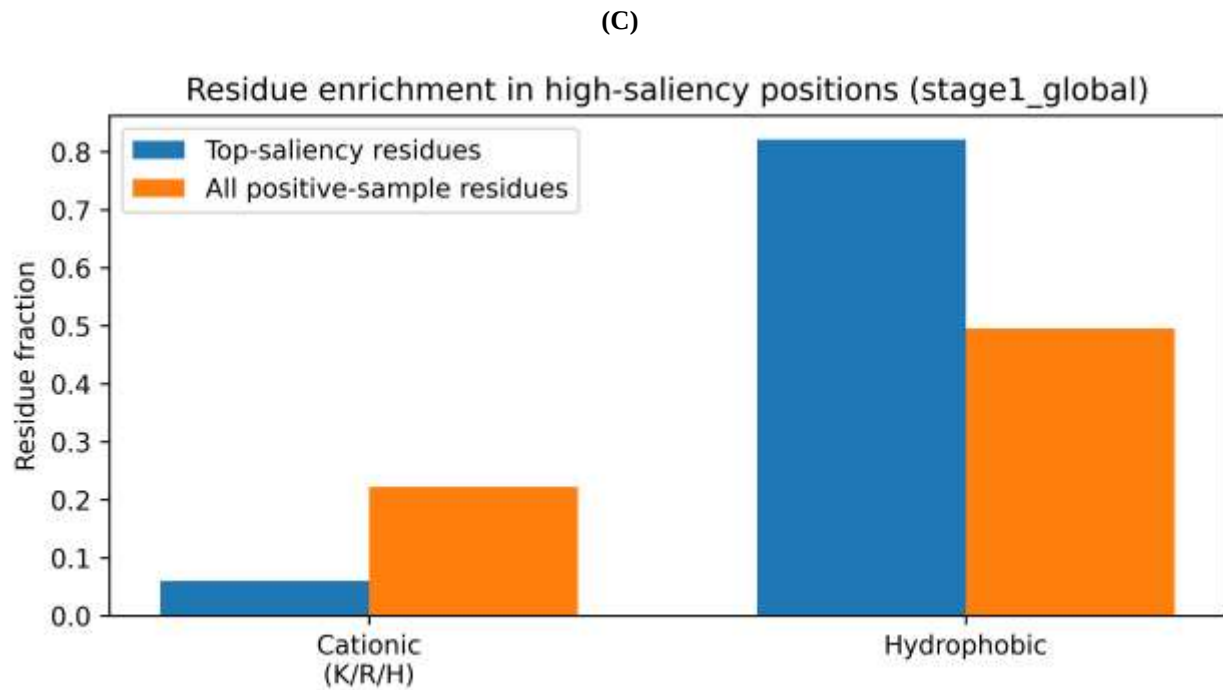


Fig. S9. Residue enrichment among high-saliency Stage-1 positions. (A) Per-amino-acid log2 enrichment relative to the background residue distribution, (B) peptide-level comparison of cationic and hydrophobic residue fractions between high-saliency positions and the corresponding peptide backgrounds, and (C) pooled residue-level enrichment of cationic and hydrophobic groups. Benjamini–Hochberg-adjusted q-values are reported in Supplementary Table S7.